

Kolmý dopad 2

Rovinná elektromagnetická vlna o kmitočtu $F=30 \cdot 10^6 \text{ Hz}$ a amplitudě $E_{im} = 100 \frac{\text{V}}{\text{m}}$

dopadá z diel. prostředí 1 $\sigma_1=0$ $\epsilon_{r1}=4$ $\mu_{r1}=1$
do diel. prostředí 2 $\sigma_2=0$ $\epsilon_{r2}=2$ $\mu_{r2}=1$

$$\epsilon_0 = 8,855 \cdot 10^{-12} \frac{\text{F}}{\text{m}} \quad \mu_0 = 4\pi \cdot 10^{-7} \frac{\text{H}}{\text{m}}$$

$$\frac{\downarrow k_i \quad \epsilon_{r1} \quad z_1}{\uparrow k_R \quad z_1} \\ \frac{\downarrow k_T \quad \epsilon_{r2} \quad z_2}{\epsilon_{r2} \quad z_2}$$

- a) Jak velký je činitel odrazu a prostupu?
b) Jak velká je amplituda elektrického a magnetického pole odražené a prostupující vlny?
c) Jak velký výkon (S_{STR}) dopadá, odraží se a prostoupí

$$a) R = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} = \frac{2 - \sqrt{2}}{2 + \sqrt{2}} = 0,172$$

$$Z_1 = \frac{Z_0}{\sqrt{\epsilon_{r1}}} = \frac{120\pi}{2} = 188,5 \Omega$$

$$Z_2 = \frac{Z_0}{\sqrt{\epsilon_{r2}}} = \frac{120\pi}{\sqrt{2}} = 266,6 \Omega$$

$$T = \frac{2Z_2}{Z_1 + Z_2} = \frac{2\sqrt{\epsilon_{r1}}}{\sqrt{\epsilon_{r2}} + \sqrt{\epsilon_{r1}}} = \frac{2 \cdot 2}{2 + \sqrt{2}} = 1,172$$

$$b) E_{im} = 100 \text{ V/m}$$

$$E_{tm} = |T| \cdot E_{im} = 117,2 \frac{\text{V}}{\text{m}}$$

$$E_{zm} = |R| \cdot E_{im} = 17,2 \frac{\text{V}}{\text{m}}$$

$$H_{im} = \frac{E_{im}}{Z_1} = \frac{100}{188,5} = 0,531 \frac{\text{A}}{\text{m}}$$

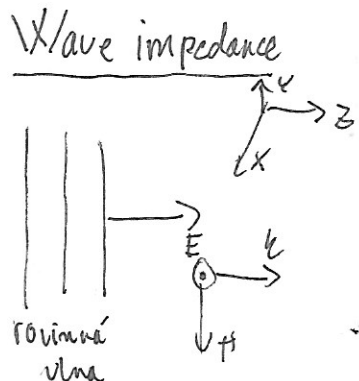
$$H_{tm} = \frac{E_{tm}}{|Z_2|} = \frac{117,2}{266,6} = 0,44 \frac{\text{A}}{\text{m}}$$

$$H_{zm} = |R| H_{im} = 0,091 \frac{\text{A}}{\text{m}}$$

$$c) S_{STR-I} = \frac{1}{2} E_{im} H_{im} = \frac{53,1}{2} = 26,55 \frac{\text{W}}{\text{m}^2}$$

$$S_{STR-R} = \frac{1}{2} E_{zm} H_{zm} = \frac{1}{2} \cdot 17,2 \cdot 0,091 = 0,783 \frac{\text{W}}{\text{m}^2} \rightarrow S_{STR-R} = |R|^2 \cdot S_{STR-I}$$

$$S_{STR-T} = \frac{1}{2} E_{tm} H_{tm} = \frac{1}{2} \cdot 117,2 \cdot 0,44 = 25,78 \frac{\text{W}}{\text{m}^2} \rightarrow$$



$$k = \sqrt{\omega^2 \mu \epsilon}$$

$$\hat{y} \left(\frac{6}{0} \right)$$

$$\vec{E} = E_0 \hat{x} e^{j(\omega t - kz)}$$

$$\vec{H} = \hat{y} H_0 e^{j(\omega t - kz)}$$

$\vec{Z}$ gives relation between $\vec{E}$ and $\vec{H}$

$$\text{rot } \vec{E} = - \frac{\partial \vec{B}}{\partial t} = - \mu \frac{\partial \vec{H}}{\partial t}$$

$$\text{rot } \vec{E} = - \mu j \omega \vec{H}$$

$$\vec{H} = \frac{\text{rot } \vec{E}}{-j \omega \mu} = \frac{-j k E_0}{-j \omega \mu} e^{j(\omega t - kz)} \hat{y}$$

$$\hat{y} H_0 e^{j(\omega t - kz)} = \frac{k}{\omega \mu} E_0 e^{j(\omega t - kz)} \hat{y}$$

$$H_0 = E_0 \sqrt{\frac{\epsilon}{\mu}} \quad E_0 = H_0 \sqrt{\frac{\mu}{\epsilon}}$$

frequency independent (free space) vlnová impedance

$$E_0 = H_0 \sqrt{\frac{\mu}{\epsilon}}$$

Pro nonmagnetic materials:

$$Z = \sqrt{\frac{\mu_0}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_0}} \left\{ \frac{1}{\sqrt{\epsilon_r}} \right\}$$

$$\frac{1}{n}$$

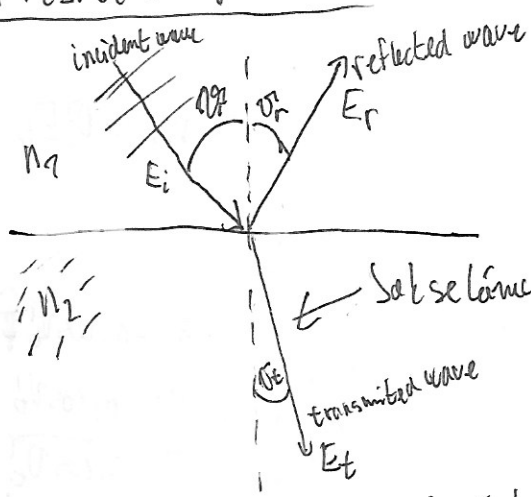
refractive index

$$\frac{Z}{Z_0} = \frac{1}{n}$$

unitless

$$Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120 \pi$$

Fresnel's equations



Snell's law

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

$$E_i = Z_0 H_i$$

Ale kolik projede / odraží se?

Odpověď: Fresnel eq.

$$E_i, E_r, E_t$$

$$n_1(E_i - E_r) = E_t + n_2$$

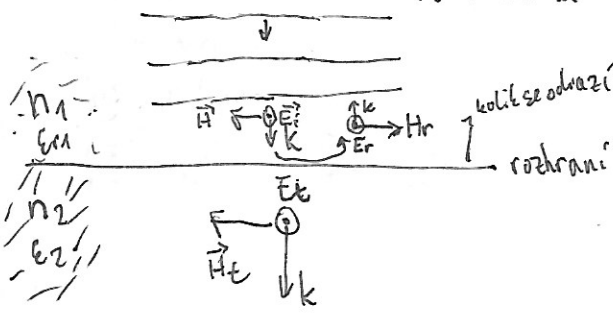
"známe Ei a Hi"

$$E_r = \frac{n_1 - n_2}{n_1 + n_2} E_i$$

$$E_t = \frac{2 n_1}{n_1 + n_2} E_i$$

$$\hat{R} = \frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$$

$$\hat{T} = \frac{2 \sqrt{\epsilon_1}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$$



co se děje na rozhraní?
 okrajové podmín.
 $E_{tani,1} = E_{tani,2}$
 Pokud na rozhr. = 0
 $H_{tani,1} = H_{tani,2}$

ale v tomto případě je
 tečná $E_{tani,1} = E_i$

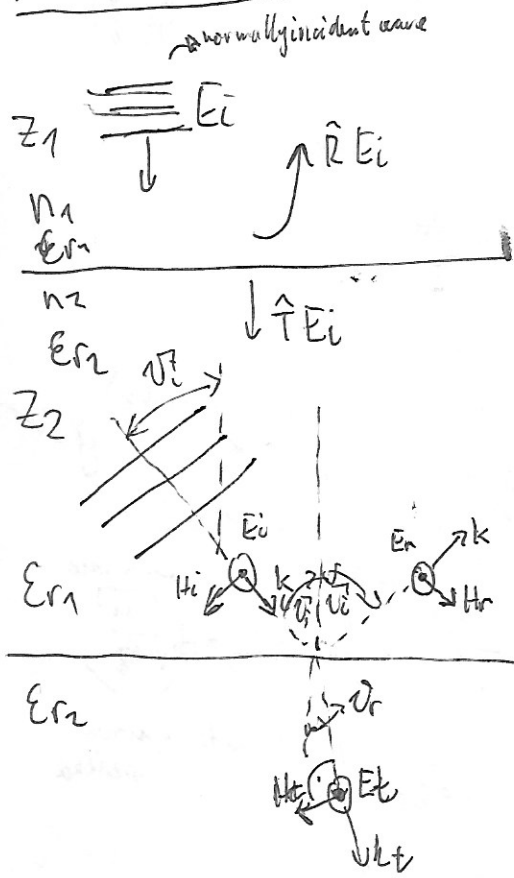
$$E_i + E_r = E_t$$

$$H_i - H_r = H_t$$

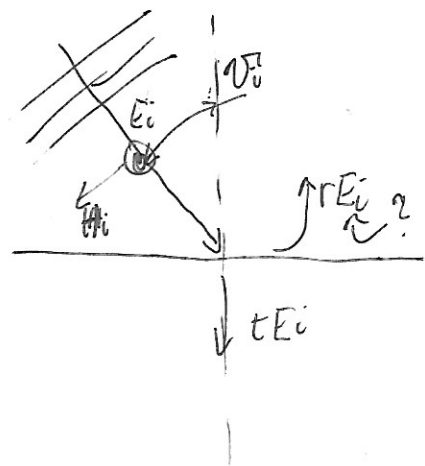
opacný směr Hr

$$E_i \quad E_r \quad E_t$$

Fresnel eq @ θ_i



obecní úhel θ_i



Polarizace pozor!

Co se tu děje? Okrajové podmínky pro záduní náhoj a proud na rozhraní

těčné složky $\begin{cases} E_{i,těčné} = E_{t,těčné} \\ H_{i,těčné} = H_{t,těčné} \end{cases}$

S polarizace
E je kolmé na popír rovinu odrazu

Co se stane s normálovou složkou H

$B_{norm,1} = B_{norm,2}$
 $H_{norm,1} = H_{norm,2}$
 $n_1 \sin \theta_i = n_2 \sin \theta_t$

① $E_i + E_r = E_t$

$H_i \cos \theta_i - H_r \cos \theta_r = H_t \cos \theta_t$

$E_i = Z H_i$

$Z = \frac{Z_0}{n} \quad n_1 \cos \theta_i (E_i - E_r) = n_2 \cos \theta_t E_t$

$E_r = \hat{R} E_i$

① $1 + R = t$

$\frac{E_r}{E_i} = R$

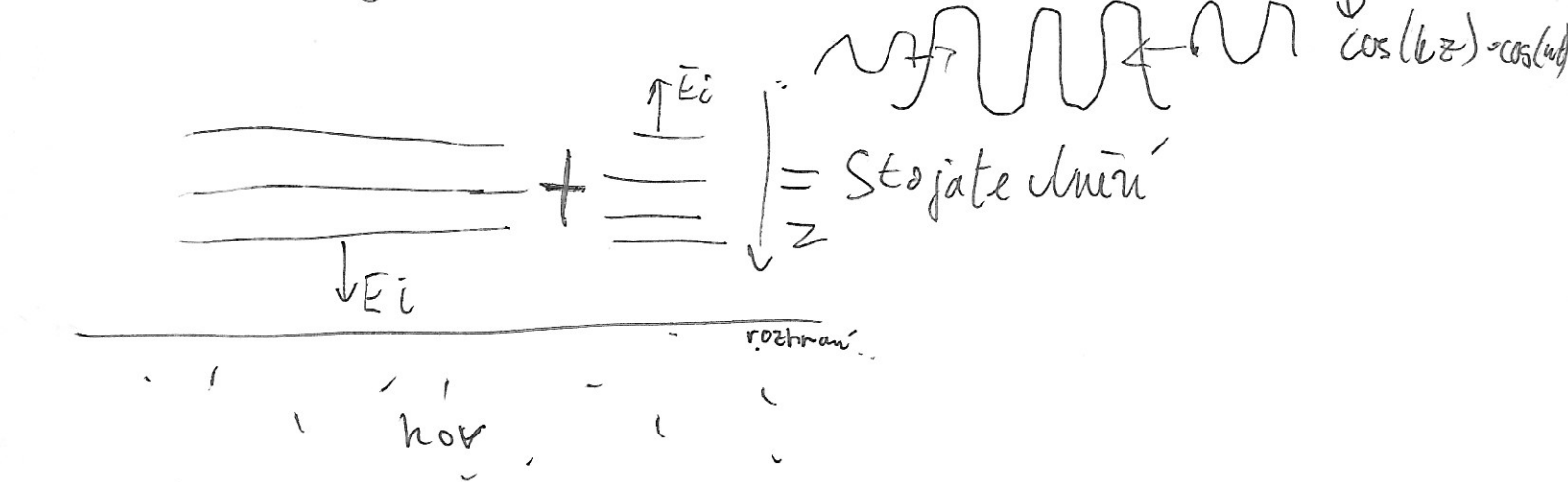
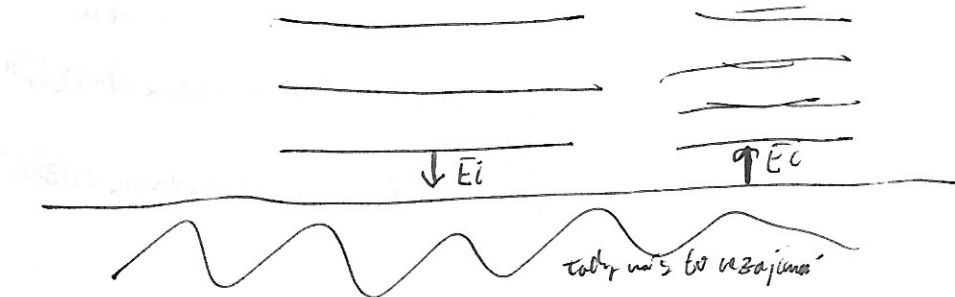
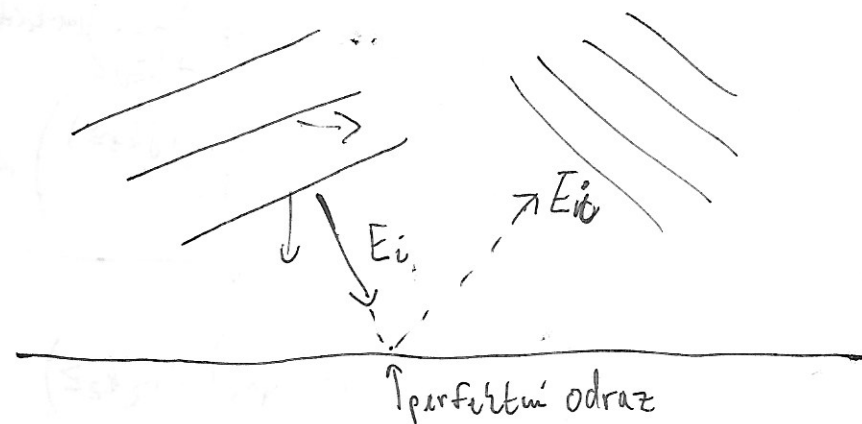
② $n_1 \cos \theta_i (1 - R) = n_2 \cos \theta_t$
 $R_s = \frac{n_1 \cos \theta_i - n_2 \cos \theta_t}{n_1 \cos \theta_i + n_2 \cos \theta_t}$

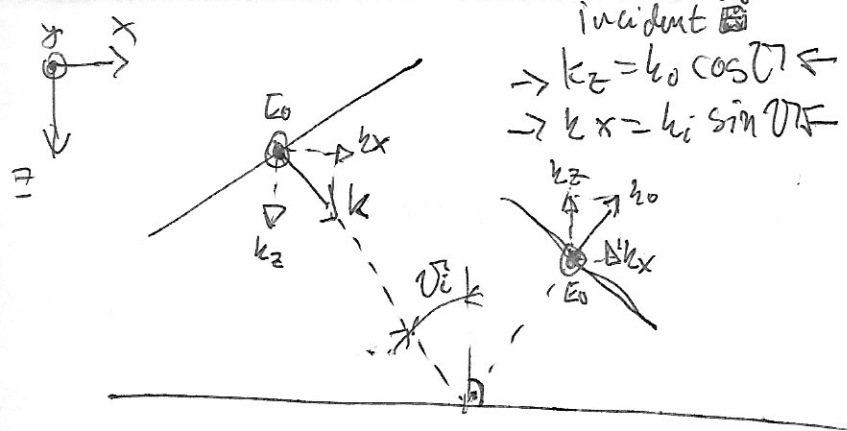
dosadím $\theta_i = 0^\circ$ a $\theta_t = 0^\circ$
 $R = \frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}$

Waveguides

~~Diffraction~~

Transverse electric TE and Transverse Magnetic TM modes





reflected

$$k_z = -k_0 \cos \theta_r$$

$$k_x = k_0 \cos \theta_r$$

$$\vec{E}_{\text{total}} = \vec{y} E_0 e^{j(\omega t - \vec{k} \cdot \vec{r})}$$

$$= \vec{y} E_0 e^{j(\omega t - k_x x - k_z z)} + \vec{y} E_0 e^{j(\omega t - k_x x + k_z z)}$$

$$= \vec{y} E_0 e^{j(\omega t - k_x x)} \left(e^{-jk_z z} + e^{jk_z z} \right)$$

$$= \vec{y} E_0 \underbrace{\cos(\omega t - k_x x)}_{\text{cestující vlna}} \underbrace{\cos(k_z z)}_{\text{stojaté vlny}}$$

$\Rightarrow$ TE mód "transverzální elektrická"

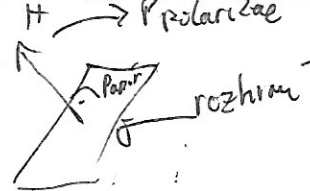
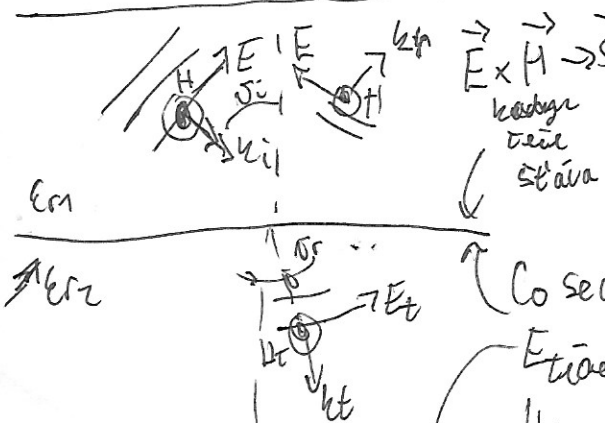
$\vec{E}$ je rovnoběžně s odrazným rozhr.

H_x a H_z

TM mód ~~je~~ je kolmé na směr šíření

TE mód E je kolmé na směr šíření

Fresnel eq. - p-polarizace



Co se děje na ohraji? (Neteče proud, není tam rákos)

$$E_{\text{tane},1} = E_{\text{tane},2}$$

$$H_{\text{tan},1} = H_{\text{tan},2}$$

$$H_i + H_r = H_t$$

$$E_i \cos \theta_i = E_r \cos \theta_i = E_t \cos \theta_t$$

Pohad je $\theta_i = 0^\circ$ tak $E_i - E_r = E_t$

$$\frac{E_i}{Z_1} + \frac{E_r}{Z_1} = \frac{E_t}{Z_2} \quad \left/ \frac{E_r}{E_i} \quad \frac{E_t}{E_i} \right.$$

cos θ

$$\hat{R} = \frac{Z_2 \cos \theta_i - Z_1 \cos \theta_t}{Z_2 \cos \theta_i + Z_1 \cos \theta_t}$$

$$\hat{T} = \frac{2 n_1 \cos \theta_i}{n_2 \cos \theta_i + n_1 \cos \theta_t}$$

$$F = 10^9 \text{ Hz} \quad \omega = 6,283 \cdot 10^9 \text{ s}^{-1}$$

$$h) S_{\text{str}}(z) = \frac{1}{2} E_m H_m e^{-2\alpha z} \cos(\varphi_z)$$

$$E_m(t=0; z=0) = 100 \frac{\text{V}}{\text{m}}$$

$$S_{\text{str}}(z=0) = 56,27$$

$$\epsilon_r = 18; \epsilon_0 = \frac{1}{36\pi} \cdot 10^{-9} \frac{\text{F}}{\text{m}}; \mu_r = 1; \mu_0 = 4\pi \cdot 10^{-7} \frac{\text{H}}{\text{m}}$$

$$S_{\text{str}}(\frac{a}{4}=z) = 55,394$$

$$\epsilon = \epsilon_r \epsilon_0 = \frac{18}{36\pi} \cdot 10^{-9} = 0,1592 \cdot 10^{-9} \frac{\text{F}}{\text{m}} \quad \mu = \mu_r \mu_0 = 12,5664 \cdot 10^{-7} \frac{\text{H}}{\text{m}}$$

$$C = 3 \cdot 10^8 \text{ m s}^{-1} \quad \sigma = 0,01 \text{ S} \quad \omega = 6,283 \cdot 10^9 \text{ s}^{-1}$$

$$a) \hat{k} = \sqrt{-j\omega\mu(j\omega\epsilon + \sigma)} = \sqrt{-j6,283 \cdot 10^9 \cdot 12,5664 \cdot 10^{-7} (j6,283 \cdot 10^9 \cdot 0,1592 \cdot 10^{-9} + 0,01)}$$

$$= \sqrt{78,975 \cdot 10^2 - j0,7895 \cdot 10^3} = 8,887$$

$$= \sqrt{+78,975 \cdot 10^2 - j0,7895 \cdot 10^3} = 88,869 - j0,444$$

$$= 88,869 - j0,444$$

$$\beta = 88,869 \text{ m}^{-1}$$

$$\alpha = 0,444 \text{ m}^{-1}$$

$$\hat{Z} = \sqrt{\frac{j\omega\mu}{j\omega\epsilon + \sigma}}$$

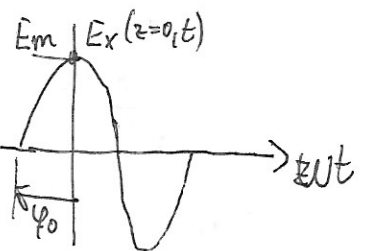
$$b) \text{ Vlnová impedanci } \hat{Z} = \frac{\hat{E}_x(z)}{\hat{H}_y(z)} = \frac{\omega\mu}{\hat{k}} = \frac{6,283 \cdot 10^9 \cdot 12,5664 \cdot 10^{-7}}{88,869 - j0,444} = (88,8 + j0,44) \Omega$$

c) Časový průběh a fázor intenzity el. pole, amplituda

$$\vec{E} = (E_x, 0, 0) \Rightarrow E_x = E_m e^{-j(\omega t - kx)}$$

$$E_x(z, t) = E_m e^{-\alpha z} \sin(\omega t - \beta z + \varphi_0) = 100 e^{-0,444z} \sin(\omega t - \beta z + \varphi_0)$$

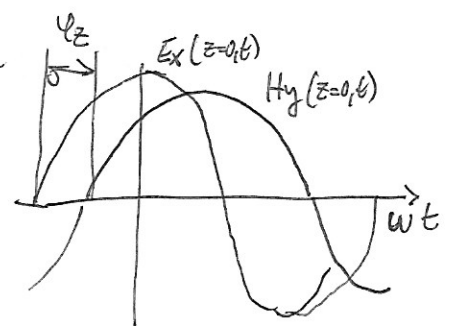
$$\hat{E}_x(z) = E_0 e^{-jkz} = \underbrace{E_m e^{j\varphi_0}}_{E_0} \underbrace{e^{-j(\beta - j\alpha)z}}_{e^{-jkz}}$$



d) Časový průběh a fázor intenzity magnetického pole, amplituda

$$H_y(z, t) = H_m e^{-\alpha z} \sin(\omega t - \beta z - \varphi_z + \varphi_0)$$

$$\hat{H}_y(z) = H_0 e^{-jkz} = \underbrace{H_m e^{j(\varphi_0 - \varphi_z)}}_{H_0} \underbrace{e^{-j(\beta - j\alpha)z}}_{e^{-jkz}}$$



$$f = 1 \cdot 10^5 \text{ Hz} \quad \omega = 2\pi f = 6,283 \cdot 10^5 \text{ s}^{-1}$$

$$\epsilon_r = 18 \quad \epsilon_0 = \frac{1}{36\pi} 10^{-9} \frac{\text{F}}{\text{m}} \quad \epsilon = \epsilon_0 \epsilon_r = 1,592 \cdot 10^{-10} \frac{\text{F}}{\text{m}}$$

$$\mu_r = 1 \quad \mu_0 = 4\pi \cdot 10^{-7} \frac{\text{H}}{\text{m}} \quad \mu = \mu_0 \mu_r = 1,257 \cdot 10^{-6} \frac{\text{H}}{\text{m}}$$

$$c = 3 \cdot 10^8 \frac{\text{m}}{\text{s}} \quad \sigma = 0,01 \frac{\text{S}}{\text{m}}$$

a) Konstanta šíření

$$\hat{k} = \sqrt{-j\omega\mu(j\omega\epsilon + \sigma)} = 0,063 - 0,063j$$

$$\beta = 0,063 \text{ m}^{-1} \cdot \text{rad}^{-1}$$

$$\alpha = 0,063 \text{ m}^{-1}$$

b) Střední hodnota $\vec{S}$ - plošná hustota činného výkonu

$$S_{\text{str}}(z) = \frac{1}{2} E_m H_m e^{-2\alpha z} \cos(\varphi_z)$$

$$S_{\text{str}}(z=0) = 400 \frac{\text{W}}{\text{m}^2}$$

$$S_{\text{str}}(z=\frac{\lambda}{4}) = 17,8 \frac{\text{W}}{\text{m}^2}$$

na vzdálenosti $\lambda/4$
poklesne S_{str} o 95,5%
 $(1 - e^{-2\alpha \cdot \frac{\lambda}{4}}) \cdot 100 = 95,5\%$

b) Vlnová impedance

$$Z = \sqrt{\frac{j\omega\mu}{j\omega\epsilon + \sigma}} = (6,314 + j6,252) \Omega$$

$$|Z| = 8,886 \Omega$$

$$\varphi_Z = 0,78 = 44,714^\circ$$

c) Časový průběh a fázor intenzity el. pole

$$E_x(z,t) = E_m e^{-\alpha z} \sin(\omega t - \beta z + \varphi_0)$$

$$\hat{E}_x(z) = E_0 e^{-j\beta z} = \underbrace{E_m e^{j\varphi_0}}_{E_0} \underbrace{e^{-j(\beta - j\alpha)z}}_{e^{-j\beta z}}$$

d) Časový průběh a fázor intenzity magn. pole

$$\hat{H}_y(z) = H_0 e^{-j\beta z} = \underbrace{H_m e^{j\varphi_0}}_{H_0} \underbrace{e^{-j(\beta - j\alpha)z}}_{e^{-j\beta z}}$$

$$H_y(z,t) = H_m e^{-\alpha z} \sin(\omega t - \beta z - \varphi_0 + \varphi_0)$$

j) Objemová hustota ztrát ve vzdálenosti $z=0$ $z=\lambda/4$

$$P(z) = \frac{1}{2} \sigma E_m^2 e^{-2\alpha z}$$

$$P(z=0) = 50 \frac{\text{W}}{\text{m}^3}$$

$$P(z=\frac{\lambda}{4}) = 2,229 \frac{\text{W}}{\text{m}^3}$$

k) Výkon na teplo

$$P = S_{\text{str}}(0) - S_{\text{str}}(\frac{\lambda}{4}) = 382 \text{ W}$$

poměr vodivosti

$$\frac{\omega\epsilon}{\sigma} = \frac{6,283 \cdot 10^5 \cdot 1,592 \cdot 10^{-10}}{0,01} = 0,01$$

Prostředí je dobré el. vodivé $\omega\epsilon \ll \sigma$
můžeme říct

$$\alpha_{\text{vodič}} = \sqrt{\frac{\omega\mu\sigma}{2}} = 0,063$$

$$\beta_{\text{vodič}} = \sqrt{\frac{\omega\mu\sigma}{2}} = 0,063$$

Pro vodivé prostředí:

$$Z_{\text{vodič}} = \sqrt{\frac{\omega\mu}{\sigma}} e^{j\frac{\pi}{4}} = (6,283 + j6,283) \Omega$$

$$|Z_{\text{vodič}}| = 8,886$$

$$\arg Z_{\text{vodič}} = \varphi_Z = 0,785 = \frac{\pi}{4} = 45^\circ$$

Fázový posuv mezi E a H je 45° ve velmi vodivém prostředí

e) Vlnová délka, fázová rychlost

$$\lambda = \frac{2\pi}{\beta} = 99,5 \text{ m}$$

$$v_f = \frac{\omega}{\beta} = 9,95 \cdot 10^6 \frac{\text{m}}{\text{s}}$$

f) Pokles amplitudy E a H na vzdálenosti $\frac{\lambda}{4}$

$$\frac{\lambda}{4} = 24,875 \quad e^{-2\alpha \frac{\lambda}{4}} \cdot 100 = 11,115\% \rightarrow \text{poklesne o } 21\%$$

$$88,885\% \rightarrow \text{poklesne o } 89\%$$

g) hloubka vniku

$$\delta = \frac{1}{\alpha} = 15,995 \text{ m}$$

Rovinná harmonická elektromagnetická vlna o kmitočtu $F = 100 \cdot 10^6 \text{ Hz}$

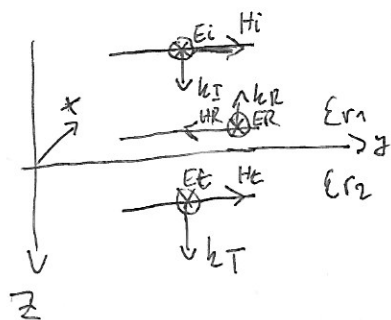
$E_{im} = 100 \frac{\text{V}}{\text{m}}$ dopadá kolmo ~~na~~ zdiel 1. na diel 2.

A) $\epsilon_{r1} = 1,5$ $\epsilon_{r2} = 8$

B) $\epsilon_{r1} = 8$ $\epsilon_{r2} = 1,5$

a) Miesta, kde rozhraním, kde budú minima uzly a maxima (kmitny)

E: Jak velká intenzita el. pole je v těchto místech?



$$\hat{E}_{xI}(z) = E_{Im} \cdot e^{-jk_I z}$$

$$\hat{E}_{xR}(z) = E_{Rm} e^{+jk_I z}$$

$$E_{Rm} = |R| \cdot E_{im} = 40 \frac{\text{V}}{\text{m}}$$

$$k = \sqrt{-j\omega\mu(j\omega\epsilon + \sigma)} = \omega \sqrt{\mu\epsilon} = 2\pi F \sqrt{\mu_0 \epsilon_0} \sqrt{\epsilon_{r1}}$$

$$k_I = 2\pi \cdot 100 \cdot 10^6 \frac{1}{3 \cdot 10^8} \sqrt{1,5} = 2,57$$

$$k_R = 2,57$$

A) $R = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} = \frac{\sqrt{1,5} - \sqrt{8}}{\sqrt{1,5} + \sqrt{8}} = -0,4$

B) $R = \frac{\sqrt{8} - \sqrt{1,5}}{\sqrt{8} + \sqrt{1,5}} = +0,4$

$$\hat{E}_{xI} + \hat{E}_{xR} = E_{Im} e^{-jk_I z} + E_{Rm} e^{+jk_I z}$$

$\Rightarrow$

$$E_{mI} \sin(\omega t - \beta_I z) + E_{mR} \sin(\omega t - \beta_I z)$$

$$= (E_{Im} + E_{Rm}) \sin(\omega t - \beta_I z)$$

A) $E_{min} = E_{im} (1 - |R|) = 100 \cdot (1 - 0,4) = 60 \frac{\text{V}}{\text{m}}$

B) $E_{max} = E_{im} (1 + |R|) = 100 (1 + 0,4) = 140 \frac{\text{V}}{\text{m}}$

0: Jelikož $\epsilon_{r1} < \epsilon_{r2}$ tak je minimum (uzel) na rozhraní a celočíselným násobkům $\frac{\lambda}{4}$ na rozhraní maxima budou v lichých násobcích $\frac{\lambda}{4}$ před rozhraním ($\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}, \dots$)

$$PSV = \frac{1 + |R|}{1 - |R|} = \frac{1,4}{0,6} = 2,3$$

$$H_{min} = H_{im} (1 - |R|) = 0,6 \cdot \frac{E_{im}}{Z_1} = \frac{0,6 \cdot 100}{\sqrt{8}} = 0,18 \frac{\text{A}}{\text{m}}$$

0: $H_{max} = H_{im} (1 + |R|) = 0,42 \frac{\text{A}}{\text{m}}$

$\epsilon_{r1} < \epsilon_{r2} : H_{min}$ -- liché násobky $\frac{\lambda}{4}$

$\epsilon_{r1} < \epsilon_{r2} : H_{max}$ -- sudé násobky $\frac{\lambda}{4}$

B) $E_{min} = E_{im} (1 - |R|) = 60 \frac{\text{V}}{\text{m}}$

$$E_{max} = 140 \frac{\text{V}}{\text{m}}$$

$\epsilon_{r1} > \epsilon_{r2}$ --- minima liché násobky $\frac{\lambda}{4}$ } před rozhraním
0: $\epsilon_{r1} > \epsilon_{r2}$ --- maxima sudé násobky $\frac{\lambda}{4}$

$$PSV = 2,3$$

0: $\epsilon_{r1} > \epsilon_{r2} : H_{min}$ -- sudé násobky $\frac{\lambda}{4}$

$\epsilon_{r1} > \epsilon_{r2} : H_{max}$ -- liché násobky $\frac{\lambda}{4}$

Rovinná harmonická elektromagnetická vlna o kmitočtu f dopadá kolmo ze vzduchu na rovinné rozhraní s dielektr. mat. Kابل rozhr. byla nalezena $E_{\max} = 150 \frac{V}{m}$ a $E_{\min} = 50 \frac{V}{m}$ vzd. mezi max. a min. je $0,4m$

- a) $\epsilon_r = ?$ ✓
- b) $E_{im} = ?$ ✓
- c) $f = ?$ ✓
- d) $PSV = ?$ ✓
- e) $S_{RT} = ?$ ✓

$$PSV = \frac{150}{50} = 3$$

$$\epsilon_{r1} = 1$$

vzduch

$$PSV = \frac{1+|R|}{1-|R|} = 3$$

$$1+|R| = 3(1-|R|)$$

$$+|R| = 3-3|R|$$

$$4|R| = 2$$

$$|R| = 0,5 \quad \epsilon_r < \epsilon_{r2}$$

$$E_{\max} = E_{im} + E_{rm} = (1+|R|)E_{im} \quad R = -0,5$$

$$E_{\min} = E_{im} - E_{rm} = (1-|R|)E_{im}$$

$$E_{im} = \frac{E_{\max}}{1+0,5} = \frac{150}{1,5} = 100 \frac{V}{m}$$

$$E_{rm} = 50 \frac{V}{m}$$

$$\lambda = \frac{c}{f}$$

$$f = \frac{c}{\lambda} = \frac{3 \cdot 10^8}{4 \cdot 0,4} = 750 MHz$$

$$T = 1 - R$$

$$T = \frac{2\sqrt{\epsilon_{r1}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} = \frac{2}{4}$$

$$R = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}}$$

$$-0,5 = \frac{1 + \sqrt{\epsilon_{r2}}}{1 + \sqrt{\epsilon_{r2}}}$$

$$\sqrt{\epsilon_{r2}} = \frac{(1+R)\sqrt{\epsilon_{r1}}}{(1-R)} = \frac{1,5}{0,5} \cdot 1 = 3$$

$$\epsilon_{r2} = 9$$

Vzdálenost mezi min a max je $\lambda/4$

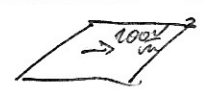
$$E_{rm} = 0,5 \cdot E_{im} = 50 \frac{V}{m}$$

$$H_{rm} = \frac{E_{rm}}{Z_2}$$

$$S_{RT} = \frac{1}{2} \frac{E_{rm}^2}{Z_2} = 0,5 \cdot \frac{50^2}{125,66} = 9,95 W/m^2$$

$$Z_2 = \frac{120\pi}{\sqrt{\epsilon_{r2}}} = 120\pi \cdot \frac{1}{3} = 40\pi = 125,66 \Omega$$

REI1 $f = 200 MHz$ $E_{mi} = 100 \frac{V}{m}$ $\sigma_2 = 57 \cdot 10^6 S/m$



a) limitová hustota proudu

$$\vec{k} = \sqrt{-j\omega\mu(j\omega\epsilon + \sigma)} = \sqrt{-j\omega\mu\sigma} = \sqrt{\frac{\omega\mu\sigma}{2}} e^{-j\frac{\pi}{4}} = \sqrt{\frac{200 \cdot 10^6 \cdot 4\pi \cdot 10^{-7}}{2}} e^{-j\frac{\pi}{4}} = 2,12 \cdot 10^5 e^{-j\frac{\pi}{4}}$$

$$j\omega\epsilon = 57 \cdot 10^6 \cdot 4\pi \cdot 10^{-12} = 7,25 \cdot 10^{-5}$$

radič
ve ϵ

$$Z_{cu} = \frac{\omega\mu}{k} = \frac{\omega\mu}{\sqrt{\frac{\omega\mu\sigma}{2}}} e^{j\frac{\pi}{4}} = \sqrt{\frac{2\omega\mu}{\sigma}} e^{j\frac{\pi}{4}} = \sqrt{\frac{2 \cdot 200 \cdot 10^6 \cdot 4\pi \cdot 10^{-7}}{57 \cdot 10^6}} e^{j\frac{\pi}{4}} = 0,0015 e^{j\frac{\pi}{4}}$$

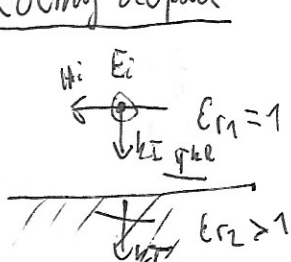
$$E_{T0} = T E_{I0} = 9,9 \cdot 10^{-4} + j 9,9 \cdot 10^{-4}$$

$$I_{cu} = \frac{\sigma_2 \cdot E_{T0}}{\sqrt{2} \cdot \sigma_2} = \frac{57 \cdot 10^6 \cdot E_{T0}}{\sqrt{2} \cdot 2,12 \cdot 10^5} = (0,19 + j 0,19) A$$

$$T = \frac{2Z_{cu}}{Z_{air} + Z_{cu}} = \frac{2 \cdot 0,0015 e^{j\frac{\pi}{4}}}{377 + j 0,0015} = \frac{0,0030 e^{j\frac{\pi}{4}}}{377 e^{-j\frac{\pi}{4}}} = 7 \cdot 10^{-6} e^{j\frac{\pi}{4}} = 9,9 \cdot 10^{-6} + j 9,9 \cdot 10^{-6}$$

$$|I_{cu}| = 0,27 A \quad I_{cu} = |H_{t0}| = \frac{|E_{T0}|}{Z_{T0}} = \frac{7 \cdot 10^{-6} e^{j\frac{\pi}{4}}}{0,0015 e^{j\frac{\pi}{4}}} = 0,00466 A/m$$

kolmý dopad



$$E_{im} = 100 \frac{V}{m} \quad f = 5 \text{ GHz}$$

$$E_{rm} = 50 \frac{V}{m}$$

$$|R| = \frac{E_{rm}}{E_{im}} = 0,5 \quad \xrightarrow{\substack{\text{výsledek} \\ \epsilon_{r1} < \epsilon_{r2}}} R = -0,5$$

$$R = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} \Rightarrow R(\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}) = \sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}$$

$$\sqrt{\epsilon_{r1}}(R+1) = \sqrt{\epsilon_{r1}}(1-R)$$

plošná hustota činného výkonu
co prostoupila

$$E_{tm} = TE_{im} = 0,5 \cdot 100 = 50 \frac{V}{m}$$

$$T = \frac{2\sqrt{\epsilon_{r1}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} = \frac{2}{1 + \sqrt{\epsilon_{r2}}} = \sqrt{\epsilon_{r1}} \frac{(1-R)}{1+R}$$

$$S_{ste-1} = \frac{1}{2} \frac{E_{tm}^2}{Z_2} \cos \varphi_z$$

1 - bezstratný materiál

$$= \frac{1}{2} \frac{50^2}{125,66} = 99,05 \frac{W}{m^2}$$

$$= 99,9 \frac{W}{m^2}$$

$$Z_2 = \frac{Z_0}{\sqrt{\epsilon_{r2}}} = \frac{120\pi}{3} = 125,66 \Omega$$

$$\epsilon_{r2} = \left(1 \cdot \frac{1+0,5}{0,5}\right)^2$$

$\epsilon_{r2} = 9$

c) Jak se zbavit odrazu?

Vložíme $\lambda/4$ vrstvu s $\sqrt{\epsilon_{r1}\epsilon_{r2}} = \sqrt{3} \Rightarrow \epsilon_{rmezí} = 3$

↑
vln. délka v antireflex vrstvě $\rightarrow \lambda_c = \frac{c}{f} \cdot \sqrt{\epsilon_{rmezí}} = \frac{3 \cdot 10^8}{5 \cdot 10^9} = \frac{3\sqrt{3}}{50}$

$$d = \frac{\lambda_c}{4} = \frac{3}{4} \sqrt{3} \cdot \frac{1}{50} = 8,66 \text{ mm}$$

Rovinná elektromagnetická vlna o kmitočtu $f = 1,2 \text{ MHz}$ dopadá kolmo ze vzduchu na vodní hladinu.

Relativní permitivita vody je $\epsilon_r = 75$, měno $\sigma = 1,5 \text{ S/m}$

$$S_{ste-2} = \frac{1}{2} \frac{E_m^2}{Z_1} \cos \varphi_z e^{-\alpha z}$$

Těsně pod hladinou σ činný výkon $S = 0,8 \frac{W}{m^2}$

$$\hat{k}_{m0} = \sqrt{-j\omega\mu(\sigma + j\omega\epsilon)} = \sqrt{-j2\pi \cdot 1,2 \cdot 10^6 \cdot 4\pi \cdot 10^{-7} \cdot (j2\pi \cdot 1,2 \cdot 10^6 \cdot 8,854 \cdot 10^{-12} \cdot 75 + 1,5)}$$

a) $= \sqrt{0,0474 - j14,2} = \sqrt{14,2 \cdot e^{-1,568j}} = 3,768 e^{-0,784j} = (2,668 - j2,66) m^{-1}$

$\arg z = 0,784 \text{ rad} = 0,8 e^{-2,66j} = 0,0039 \frac{W}{m^2}$

$$\hat{Z}_{m0} = \frac{\omega\mu}{\hat{k}} = \frac{2\pi \cdot 1,2 \cdot 10^6 \cdot 1,4 \cdot 10^{-7}}{2,668 - j2,660} = 2,515 e^{j0,784} \Omega = (1,781 + j1,776) \Omega$$

$E_{em}(z=0) = \sqrt{2 \cdot S \cdot Z} \cos \varphi_z = \sqrt{2 \cdot 0,8 \cdot 2,515} (\cos(0,784)) = 2,384 \frac{V}{m}$

$E_{rm}(z=1) = E_{rm}(0) \cdot e^{-\alpha z} = 2,384 e^{-2,66} = 1,668 \frac{V}{m}$ $H_{rm}(z=0) = 0,069 \frac{A}{m}$ $H_{rm}(z=1) = 0,069 \frac{A}{m}$

b) $E_{Rm}(z=0) = E_{im} + E_{rm}$

$$E_{Rm}(z=0) = \frac{375,21}{378,776} e^{j1,771} = 0,99 e^{j1,771} = -0,196 + j0,97$$

$$T = \frac{2\sqrt{\epsilon_{r1}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} = \frac{2 \cdot 1}{1 + \sqrt{75}} = 0,121$$

$$R = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{1,781 + j1,776 - 120\pi}{1,781 + j1,776 + 120\pi} = \frac{-375,21 + j1,776}{378,776 e^{j0,00469}} = \frac{-375,21 + j1,776}{378,776 e^{j0,00469}}$$

$$V = \frac{Z_2}{Z_1} = \frac{2 \cdot 2,515 e^{j0,784}}{1,781 + j1,776} = \frac{5,03 e^{j0,784}}{378,776 e^{j0,00469}} = \frac{5,03 e^{j0,784}}{378,776 e^{j0,00469}} = 0,013 e^{j0,779}$$

Kolmý dopad 2

Rovinná elektromagnetická vlna o kmitočtu $F=30 \cdot 10^6 \text{ Hz}$ a amplitudě $E_{im} = 100 \frac{\text{V}}{\text{m}}$

dopadá z diel. prostředí 1 $\sigma_1=0$ $\epsilon_{r1}=4$ $\mu_{r1}=1$
do diel. prostředí 2 $\sigma_2=0$ $\epsilon_{r2}=2$ $\mu_{r2}=1$

$$\epsilon_0 = 8,855 \cdot 10^{-12} \frac{\text{F}}{\text{m}} \quad \mu_0 = 4\pi \cdot 10^{-7} \frac{\text{H}}{\text{m}}$$

$$\frac{\downarrow k_{zi} \quad \epsilon_{r1} \quad z_1}{\uparrow k_{zr} \quad z_1} \\ \frac{\downarrow k_{zt} \quad \epsilon_{r2} \quad z_2}{\uparrow k_{zr} \quad z_2}$$

- Jak velký je činitel odrazu a prostupu?
- Jak velká je amplituda elektrického a magnetického pole odražené a prostupující vlny?
- Jak velký výkon (S_{STR}) dopadá, odraží se a prostupí

$$a) R = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} = \frac{2 - \sqrt{2}}{2 + \sqrt{2}} = 0,172$$

$$Z_1 = \frac{Z_0}{\sqrt{\epsilon_{r1}}} = \frac{120\pi}{2} = 188,5 \Omega$$

$$Z_2 = \frac{Z_0}{\sqrt{\epsilon_{r2}}} = \frac{120\pi}{\sqrt{2}} = 266,6 \Omega$$

$$T = \frac{2Z_2}{Z_1 + Z_2} = \frac{2\sqrt{\epsilon_{r1}}}{\sqrt{\epsilon_{r2}} + \sqrt{\epsilon_{r1}}} = \frac{2 \cdot 2}{2 + \sqrt{2}} = 1,172$$

$$b) E_{im} = 100 \text{ V/m}$$

$$E_{tm} = |T| \cdot E_{im} = 117,2 \frac{\text{V}}{\text{m}}$$

$$E_{rm} = |R| \cdot E_{im} = 17,2 \frac{\text{V}}{\text{m}}$$

$$H_{im} = \frac{E_{im}}{Z_1} = \frac{100}{188,5} = 0,531 \frac{\text{A}}{\text{m}}$$

$$H_{tm} = \frac{E_{tm}}{|Z_2|} = \frac{117,2}{266,6} = 0,44 \frac{\text{A}}{\text{m}}$$

$$H_{rm} = |R| H_{im} = 0,091 \frac{\text{A}}{\text{m}}$$

$$c) S_{STR-I} = \frac{1}{2} E_{im} H_{im} = \frac{53,1}{2} = 26,55 \frac{\text{W}}{\text{m}^2}$$

$$S_{STR-R} = \frac{1}{2} E_{rm} H_{rm} = \frac{1}{2} \cdot 17,2 \cdot 0,091 = 0,783 \frac{\text{W}}{\text{m}^2} \rightarrow S_{STR-R} = |R|^2 \cdot S_{STR-I}$$

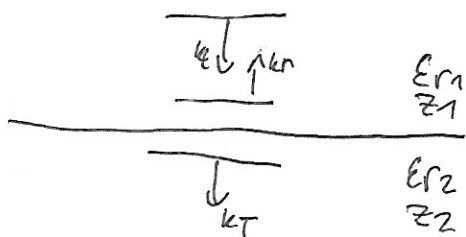
$$S_{STR-T} = \frac{1}{2} E_{tm} H_{tm} = \frac{1}{2} \cdot 117,2 \cdot 0,44 = 25,78 \frac{\text{W}}{\text{m}^2} \rightarrow$$

Rovinná elektromagnetická vlna o frekvenci $F = 2 \text{ GHz}$ dopadá kolmo na rozhraní vakuum-dielektrikum. Relativní permitivita dielektrika je rovna 4. Maximální hodnota intenzity elektrického pole dopadající vlny je 1 kV/m . Určete:

- Maximální hodnotu intenzity elektrického pole v dielektriku
- Časovou střední hodnotu plošné hustoty výkonu neseného vlnou v dielektriku
- kolik procent výkonu se odrazí
- Vlnovou délku vlny v dielektriku

$$\hat{k}_I = \hat{k}_R =$$

$$\hat{k}_T =$$



a) Z okrajových podmínek víme

$$T = \frac{2n_1}{n_1 + n_2} = \frac{2\sqrt{\epsilon_{r1}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} = \frac{2}{1+2} = \frac{2}{3}$$

$$E_{mT} = T \cdot E_{mI} = \frac{2}{3} \cdot 1000 = 666.7 \text{ V/m}$$

b) $S_{\text{stř}} = \frac{1}{2} E_m e^{-\frac{1}{2}z} H_m e^{-\frac{1}{2}z} \cos(\varphi_{zT}) = \frac{1}{2} E_m \cdot H_m = \frac{1}{2} \cdot 666.7 \cdot 3.54 = 1180 \frac{\text{W}}{\text{m}^2}$

$$Z_T = \frac{E_m}{H_m} = Z_1 = Z_2 = \frac{120\pi}{\sqrt{\epsilon_{r2}}} = \frac{120\pi}{2} = 60\pi = 188.5 \Omega$$

$\varphi_z = 0$

$$H_{mT} = \frac{E_{mT}}{Z_T} = \frac{666.7}{188.5} = 3.54 \text{ A/m}$$

$k = \sqrt{\omega^2 \mu \epsilon} = \omega \sqrt{\mu_0 \epsilon_0 \epsilon_r} = 2\pi F \frac{1}{c} \sqrt{\epsilon_r} = 2\pi \frac{2 \cdot 10^9}{3 \cdot 10^8} \cdot 2 = \frac{80}{3} \pi = \beta$

$\mu_0 = 1$
 $\epsilon_0 = 1$

d) $\lambda = \frac{2\pi}{\beta} = \frac{6}{80} \text{ m} = \frac{3}{40} = 0.075 \text{ m}$

c) $R = \frac{Z_2 - Z_1}{Z_1 + Z_2} = \frac{1-2}{1+2} = -\frac{1}{3}$

$|R|^2 = \frac{1}{9}$

$$R = \frac{Z_2 - Z_1}{Z_1 + Z_2} = \frac{\sqrt{\epsilon_{r1}} - \sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}}$$

Odrazí se $\frac{1}{9}$ výkonu

REM $F = 200 \text{ MHz}$ $E_{im} = 100 \frac{\text{V}}{\text{m}}$ $G_2 = 57 \cdot 10^6 \text{ S/m}$

$$Z_2 = \frac{\omega \mu}{\hat{k}} = \frac{2\pi \cdot 2 \cdot 10^8 \cdot 4\pi \cdot 10^{-7}}{\sqrt{2\pi \cdot 2 \cdot 10^8 \cdot 4\pi \cdot 10^{-7} \cdot 57 \cdot 10^6}} e^{j\frac{\pi}{4}}$$

$$\hat{k} = \sqrt{-j\omega\mu\sigma}$$

$$\hat{k} = e^{-\frac{\pi}{4}j} \sqrt{\omega\mu\sigma}$$

$$= 0,005 e^{j\frac{\pi}{4}} = (0,0037 + j0,0037) \Omega$$

$$\hat{T} = \frac{2Z_2}{Z_1 + Z_2} = \frac{0,01 e^{j\frac{\pi}{4}}}{120\pi + 0,0037 + j0,0037} = \frac{0,01 e^{j\frac{\pi}{4}}}{376,995 + j0,0037} = \frac{0,01 e^{j\frac{\pi}{4}}}{376,995 e^{j9,82 \cdot 10^{-6}}} =$$

$$= 0,000027 e^{j\frac{\pi}{4}} = 0,0000188 + j0,0000188$$

$$E_{Tm_0} = \hat{T} E_{im} = 0,0027 e^{j\frac{\pi}{4}} =$$

a) $|\hat{H}_{Tm_0}| = \left| \frac{\hat{E}_{Tm_0}}{\hat{Z}_2} \right| = \frac{0,0027 e^{j\frac{\pi}{4}}}{0,005 e^{j\frac{\pi}{4}}} = 0,54 \frac{\text{A}}{\text{m}}$

b) $S_t = \frac{1}{2} E_{tm} H_{tm} \cos \frac{\pi}{4} = \frac{1}{2} \cdot 0,0027 \cdot 0,54 \cos \frac{\pi}{4} = 0,0005 \frac{\text{W}}{\text{m}^2}$

6)

$$\epsilon_{r1} = 1$$

Budu předpokládat, že se jedná o $\lambda/4$ vrstvu na eliminaci odrazu

$$\epsilon_{r \text{ vrstva}} = \sqrt{\epsilon_{r1} \epsilon_{r2}} = \sqrt{1 \cdot 4} = 2$$

$$\frac{\lambda_{\text{vrstva}}}{4} = 5 \text{ mm} \rightarrow \lambda_{\text{vrstva}} = 20 \text{ mm}$$

$$f = \frac{c}{\lambda} = \frac{c}{\lambda_{\text{vrstva}}} = \frac{c}{20 \text{ mm}}$$

Bez vrstvy

$$T = \frac{\sqrt{\epsilon_{r1}} \cdot 2}{\sqrt{\epsilon_{r1}} + \sqrt{\epsilon_{r2}}} = \frac{2}{3}$$

$$|T|^2 = \frac{4}{9}$$

$$f = \frac{3 \cdot 10^8}{0,02 \sqrt{2}} = 10,6 \cdot 10^9$$

$$= 10,6 \text{ GHz}$$

$$Z_2 = \frac{Z_0}{\sqrt{\epsilon_{r2}}}$$

$$S_{\text{bez vrstvy}} = \frac{4}{9} \cdot 0,5 = 0,444 \frac{\text{W}}{\text{m}^2}$$

$$S_{\text{bez vrstvy}}(z = -5) = |T|^2 \cdot \frac{\sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}}} S_{\text{vrstva}} = \frac{4}{9} \cdot 2 \cdot 0,5 = 0,444 \frac{\text{W}}{\text{m}^2}$$

$$S_{T_{\text{ref}}} = |T|^2 \frac{Z_1}{Z_2} S_{I_{\text{ref}}} = |T|^2 \cdot \frac{\sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}}} \frac{Z_0}{Z_0} = \frac{\sqrt{\epsilon_{r2}}}{\sqrt{\epsilon_{r1}}} |T|^2 S_{I_{\text{ref}}}$$

